

MIN-MAX ± 1 QUADRATIC FORM

Partial results toward MO 413935 · existence of the limit still OPEN

THE QUESTION

Does the following limit exist?

$$L = \lim_{n \rightarrow \infty} n^{-3/2} \min_{a_{ij} = \pm 1} \max_{x_i = \pm 1} \left| \sum_{i < j} a_{ij} x_i x_j \right|$$

Write $\alpha_n = m_n/n^{3/2}$ with m_n the min-max above.

PROVED — SANDWICH

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \alpha_n \leq \limsup_{n \rightarrow \infty} \alpha_n \leq \frac{1}{2}$$

Lower: dual-Gaussian arcsine (every Seidel matrix).

Upper: Paley / conference matrices + denseness of Paley orders.

EQUIVALENT FORMS

$\sum_{i < j} a_{ij} x_i x_j = \frac{1}{2} x^\top A x$ for symmetric zero-diagonal Seidel A .

$$m_n = \min_A \Phi(A), \quad \Phi(A) = \max_x \left| \frac{1}{2} x^\top A x \right| = \frac{1}{2} n \|A\|_{\text{op}} \rho(A)$$

Cut-code: $m_n = \binom{n}{2} - 2 \rho(D_n)$ with $D_n = \{\pm(x_i x_j)_{i < j}\}$.

STATUS

Existence of $L = \lim \alpha_n$ is OPEN — neither proved nor disproved.

The sandwich alone does not force $\liminf = \limsup$.

WHAT'S NEW · WHAT'S OPEN

$\rho = 1$ on a dense Paley family · exact small- n table · blockers for existence

THEOREM — $\rho = 1$ FOR PALEY ORDER p^2+1

For odd prime p , Paley conference C over F_{p^2} ($n = p^2 + 1$) has a halfspace boolean eigenvector:

$$Cx = \rho x, \quad x \in \{\pm 1\}^n \Rightarrow \rho(C) = 1, \quad \Phi(C) = \frac{1}{2} n \sqrt{n-1}$$

Along $n_k = p_k^2 + 1$: $n_{k+1}/n_k \rightarrow 1$ and $\limsup \rho = 1$. Checked for $p = 3, 5, 7$ ($n = 10, 26, 50$).

EXACT m_n (CERTIFIED)

n	2	3	4	5	6	7	8	9	10
m_n	1	3	4	4	5	9	10	12	13

At $n = 10$ (Paley $p = 3$): $m_{10} = 13 < \Phi_{\text{Paley}} = 15$. Conference is not exactly optimal — E(1) must be asymptotic

OPEN BLOCKERS (FOR EXISTENCE OF L)

- ▶ E(1) Asymptotic optimality: $m_{n_k} = \Phi(C_k) + o(n_k^{3/2})$ along Paley.
- ▶ E(2) $\rho(C_k) \rightarrow \rho_*$ for all Paley (already $\rho = 1$ on $p^2 + 1$).
- ▶ Or: two subsequences with unequal $\lim \alpha$ (non-existence).
- ▶ Soft multipartite / Hadamard / Q4 / SA local- ϕ alone cannot finish this.

CONDITIONAL PICTURE

If E(1)+E(2): $L = \rho_*/2$. On the $\rho = 1$ family, E(1) alone would give $\limsup \alpha = 1/2$;

matching $\liminf = 1/2$ still needs more than dual-Gauss $1/\pi$. Neither is proved.